

V_0 : Raw Verifier Output

```
(((((... ong long) 57 * b) + sum) ==  
((long lon... && 57 == i  
  || long long) b * 54) + sum) ==  
((long lon... && i == 54  
  || long long) b * 36) + sum) ==  
((long lon... && i == 36  
  || 5 == i && long long) 5 * a) ==  
(sum + ((long long...  
:  
:  
  || long long) 2 * b) + sum) ==  
((long long... && 2 == i
```

7,476 chars, 100 disjuncts

V_1 : Normalized

```
57 * b + sum == 57 * a && 57  
== i  
  || b * 54 + sum == a * 54  
&& i == 54  
  || b * 36 + sum == a * 36  
&& i == 36  
  || 5 == i && 5 * a == sum +  
5 * b  
:  
:  
  || 2 * b + sum == 2 * a &&  
2 == i
```

3,686 chars, 100 disjuncts

normalize

LLM

simplify

V_2 : Simplified

```
sum == (a - b) * i
```

18 chars, 1 predicate